
Toward Effective Multimodal Graph Foundation Model: A Divide-and-Conquer Based Approach

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Abstract

Graph Foundation Models (GFMs) have achieved remarkable success in generalizing across diverse domains. However, they mainly focus on Text-Attributed Graphs (TAGs), leaving Multimodal-Attributed Graphs (MAGs) largely untapped. Although recent Multimodal Graph Foundation Models (MGFMs) attempt to bridge this gap by integrating diverse modality information, our empirical investigation reveals two fundamental limitations: ❶ they fail to explicitly model modality interaction, essential for capturing intricate cross-modal semantics beyond simple aggregation, and ❷ they exhibit sub-optimal modality alignment, which is critical for bridging the significant semantic disparity between distinct modal spaces. To address these challenges, we propose **PLANET** (graPh topoLology-aware modAlity iNteraction and alignmEnT), a novel framework employing a Divide-and-Conquer strategy to decouple modality interaction and alignment across distinct granularities. At the embedding granularity, ❶ Embedding-wise Domain Gating (EDG) performs local semantic enrichment by adaptively infusing topology-aware cross-modal context, achieving modality interaction. At the node granularity, ❷ Node-wise Discretization Retrieval (NDR) ensures global modality alignment by constructing a Discretized Semantic Representation Space (DSRS) to bridge modality gaps. Extensive experiments demonstrate that PLANET significantly outperforms state-of-the-art baselines across diverse graph-centric and multimodal generative tasks. Our code is available at <https://github.com/Sapnapnet/PLANET>.

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1. Introduction

In recent years, GFMs (Xia et al., 2024; Xia & Huang, 2024) have emerged as a transformative paradigm in graph representation learning, offering a unified encoding framework capable of generalizing across cross-domain datasets. However, most existing GFMs are primarily designed for TAGs (He et al., 2025a; Wang et al., 2024b) or focus on learning unified graph structures (Yu et al., 2025; Sun et al., 2025), failing to extend effectively to MAGs (Zhu et al., 2025a; Yan et al., 2025). This limitation implies significant missed improvements. From a data perspective, MAGs enriched with diverse modalities offer significantly richer multimodal semantic information compared to TAGs. From an application perspective, extending GFMs to MAGs significantly broadens the applicable downstream tasks such as modality retrieval task (Qu et al., 2021) and graph generative task (Yoon et al., 2023; Fang et al., 2025). To address this limitation, MGFMs (He et al., 2025b; Fang et al., 2025) have been introduced to integrate these heterogeneous signals within graph structures, achieve domain and modality alignment.

To validate the advantages of incorporating multimodal information and the superiority of MGFMs approaches, we conducted an empirical study shown in Fig. 1(a)&(b). Our findings yield two key conclusions: ❶ The use of multiple modalities consistently improve model performance. ❷ MGFMs significantly outperform GFMs. UniGraph2 (He et al., 2025b) consistently achieves state-of-the-art results, demonstrating the need for specialized architectures for MAGs. Please refer to Appendix B.1 for detailed implementation.

Although MGFMs such as UniGraph2 have demonstrated a powerful capacity to learn from diverse multimodal graphs, they are fundamentally limited by their approach to modality interaction and alignment. Specifically, UniGraph2 does not account for the significant semantic disparities between the latent spaces of different encoders and, crucially, lacks modality interaction, which is verified to be important in learning robust feature representations (Zhu et al., 2025a). Furthermore, UniGraph2 lacks explicit supervisory signals for modality alignment, rendering the alignment process inefficient and inadequately constrained.

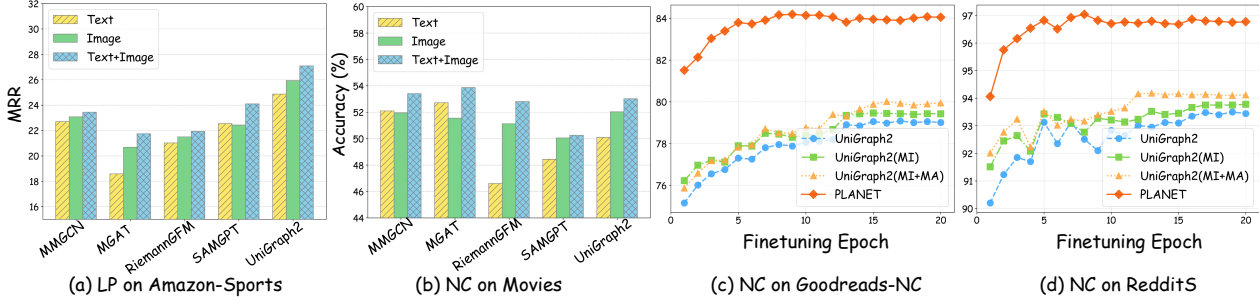


Figure 1. Empirical study results. (a)&(b) Performance comparison across different modalities and architectures. (c)&(d) Stepwise enhancement results.

To empirically substantiate these limitations of UniGraph2, we consider the general architecture of MGFM, which typically consists of four core components: ❶ Modality-specific encoding, ❷ Modality Interaction (including Graph Neural Network (GNN) processing), ❸ Modality Alignment, and ❹ Modality Fusion. As detailed in our ablation studies (See Fig. 1(c)&(d)), we involve a stepwise enhancement of the UniGraph2 architecture, progressively endowing it with capabilities for Modality Interaction (MI) and Modality Alignment (MA). The results demonstrate that each module provides a positive contribution to the model’s overall performance. This analysis forms the primary motivation for our work: to develop dedicated mechanisms for these two critical modules, tailored to address their challenges at distinct granularities. Please refer to Appendix B.2 for detailed implementation of the empirical study.

Motivated by the limitations mentioned above, in this work, we propose PLANET, a novel framework that utilizes a Divide-and-Conquer strategy to decouple the complexity inherent in MGFM design across two distinct granularities. At the embedding granularity, we introduce the ❶ **EDG for Modality Interaction**. We formulate modality interaction as a process of local semantic enrichment. This module employs a domain-specialized gating mechanism to adaptively extract and infuse topology-aware cross-modal context, enhancing the model’s comprehension of intricate inter-modality relationships at the fine-grained embedding level. At the node granularity, we propose the ❷ **NDR for Modality Alignment**. We view modality alignment as a global semantic consensus problem. By constructing a Discretized Semantic Representation Space (DSRS), this module retrieves and anchors heterogeneous signals into a unified space, explicitly enforcing robust alignment at the coarse-grained node level.

In summary, our **key contributions** in PLANET are: (1) *New Perspective*. To the best of our knowledge, we are the first to systematically identify and address the critical shortcomings of existing MGFM with respect to lack of modality interaction and alignment. (2) *Novel Approach*.

We propose PLANET, a novel framework that introduces a paradigm of topology-aware modality interaction and modality alignment, offering a valuable reference for future designs in the MGFM domain. (3) *SOTA Performance*. Extensive experiments demonstrate the superiority of PLANET across various graph-centric tasks and multimodal generative tasks.

2. Preliminaries

2.1. Notations and Problem Formulation

Consider a MAG denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, with $|\mathcal{V}| = N$ nodes and $|\mathcal{E}| = M$ edges. $\mathcal{R}$ denotes the collection of raw multimodal data associated with the nodes (e.g., raw text descriptions, images). Let $\mathcal{M} = \{m_1, m_2, \dots, m_{|\Omega|}\}$ be the set of available modalities, where $|\Omega|$ denotes the number of modalities. In this work, we operate under the assumption that each node $v \in \mathcal{V}$ possesses complete features across all modalities in $\mathcal{M}$, ensuring a comprehensive multimodal context for every entity. The raw data is transformed into feature $\mathbf{X}^{(m)} \in \mathbb{R}^{N \times d_m}$ via ϕ_m , where ϕ_m denotes the frozen modality-specific encoders for modality m (e.g., ViT (Dosovitskiy, 2020) for images, BERT (Devlin et al., 2019) for texts). Consequently, we obtain a feature-transformed graph $\mathcal{G}' = (\mathcal{V}, \mathcal{E}, \mathcal{X})$, where $\mathcal{X} = \{\mathbf{X}^{(m)}\}_{m \in \mathcal{M}}$ serves as the input node features for the subsequent learning process.

Based on $\mathcal{G}'$, we aim to learn a unified embedding function f_θ following the standard paradigm: **Pre-training stage**. We optimize f_θ using self-supervised objectives (e.g., reconstruction tasks and contrastive tasks). **Fine-tuning stage**. The pre-trained f_θ is adapted with task-specific heads to support diverse downstream tasks.

2.2. Multimodal Graph Learning

Multimodal Graph Learning (MGL) is the machine learning technique on multimodal graphs (Peng et al., 2024). MGL are widely employed across various domains, including biology (Gainza et al., 2020), chemistry (Guan et al., 2021),

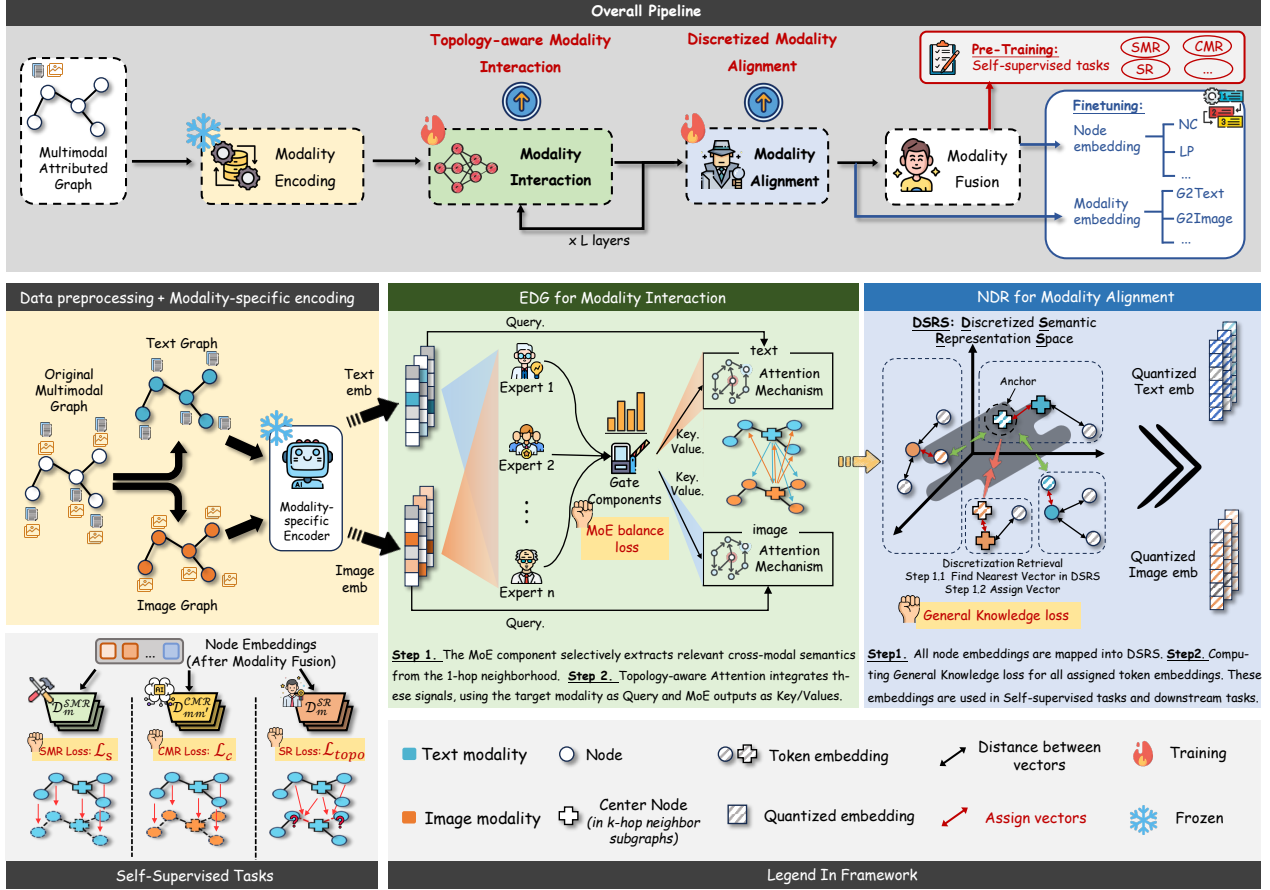


Figure 2. Overall architecture of the proposed method: PLANET.

knowledge graph (Chen et al., 2022; Zeng et al., 2023), healthcare applications (Zheng et al., 2022) and recommendation systems (Wei et al., 2019; Tao et al., 2020; Yi et al., 2022; Zhang et al., 2021). However, these existing models are inherently designed for specific tasks within specific domains (e.g., user-item link predictions), lacking the generalization capability to transfer effectively to other domains or downstream tasks.

2.3. Graph Foundation Models

Recently, Graph Foundation Models (GFMs) have gained significant attention for their robust generalization capabilities (Liu et al., 2025b). Some approaches focused on capturing universal structural patterns (Chen et al., 2025; Yu et al., 2025; Sun et al., 2025). The majority of existing GFMs concentrate on TAGs, which focus on aligning graph structural representations with the semantic space (Tang et al., 2024; Chen et al., 2024; Kong et al., 2025; Zhu et al., 2025b; Li et al., 2024) and unifying diverse downstream tasks across domains to achieve transferability (Liu et al., 2024; Huang et al., 2023). However, these GFMs are not

primarily designed for MAGs.

Unlike previous works, UniGraph2 (He et al., 2025b) establishes a MGFM, but fails to effectively address the critical issues of modality interaction and alignment (see Sec. 1). Although a new study GraphGPT-O (Fang et al., 2025) attempts to incorporate cross-modal interaction, it employs a non-topology-aware modality interaction mechanism within an LLM-based framework restricted to generative tasks, thereby suffering from prohibitive computational overhead and limited versatility.

3. Methodology

3.1. Overview

We propose **PLANET**, a MGFM that uses a Divide-and-Conquer strategy to decouple modality interaction and modality alignment (Fig. 2).

Modality Encoding. We use modality-specific encoders to transform raw multimodal data into feature vectors. Following Hou et al. (2022), we employ a modality masking

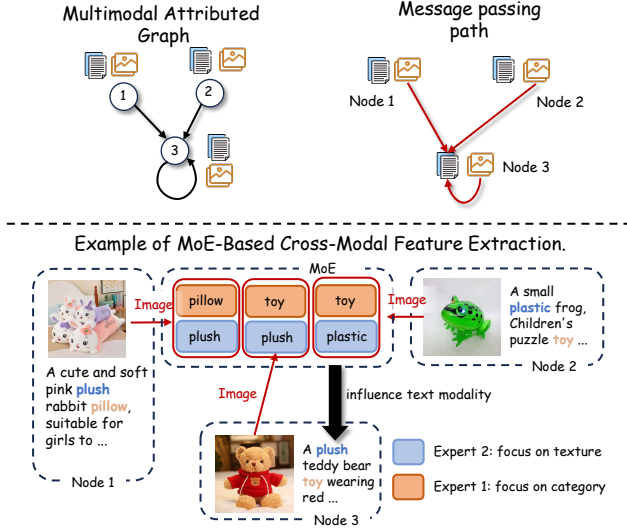


Figure 3. Illustration of expert-driven semantic extraction within the EDG module. Taking the text modality as the target example, we observe that the target text modality (in Node 3) correlates with neighboring images (in Node 1,2) via diverse attributes. Our MoE module is designed to capture these distinct semantic patterns through specialized experts, enabling the precise extraction of effective cross-modal mutual information.

strategy to these feature vectors (details in Appendix C.1), yielding the masked feature vectors $\tilde{\mathbf{X}}^{(m)}$.

For feature dimension alignment across heterogeneous modalities, we apply a set of modality-specific MLPs on $\tilde{\mathbf{X}}^{(m)}$: $\mathbf{H}^{(0,m)} = \text{MLP}_m(\tilde{\mathbf{X}}^{(m)})$. Subsequently, to preserve the inherent distribution of each modality, we process $\mathbf{H}^{(0,m)}$ through independent, modality-specific GNNs (e.g., Graph Transformer (Dwivedi & Bresson, 2020)) to obtain the modality-specific embeddings $\mathbf{H}^{(spe,m)}$.

Modality Interaction & Modality Alignment. We first employ the EDG at the embedding granularity to infuse topology-aware cross-modal context (Sec. 3.2). Subsequently, the NDR operates at the node granularity, anchoring features into a DSRS for modality alignment. This process yields the aligned modality-level node representations $\mathbf{H}^{(cross,m)}$ (Sec. 3.3).

Modality Fusion. To integrate modality specificity with interactive semantics, we first fuse modality embeddings: $\mathbf{H}^{(all,m)} = \mathbf{H}^{(spe,m)} \parallel \mathbf{H}^{(cross,m)}$. Specifically, for graph-centric tasks (e.g., node classification), we further concatenate multimodal representations to obtain the final node embeddings: $\mathbf{h}_i = \parallel_{m \in \mathcal{M}} \mathbf{h}_i^{(all,m)}$. The entire model is optimized via a joint self-supervised objective (Sec. 3.4).

3.2. EDG for Modality Interaction

Following the Divide-and-Conquer strategy, the first challenge is to model modality interaction. We formulate this

as a process of local semantic enrichment at the embedding granularity. Unlike previous works that aggregate modalities globally (He et al., 2025b), EDG operates within the layer-wise propagation, allowing each node’s modality embedding to dynamically absorb complementary semantics from its neighbors before any global alignment is enforced.

MoE-Based Cross-Modal Feature Extraction. Intuitively, in a MAG, the semantic representation of a node in one modality (e.g., text) is often correlated with the complementary modalities (e.g., image) of its neighbors and itself. For example, a textual description may correlate with the texture and category of objects in images (See Fig. 3). Guided by this observation, we employ a Mixture-of-Experts (MoE) module to capture these diverse semantic patterns. Formally, taking $\mathbf{H}^{(0,m)}$ as input, let $\mathbf{h}_i^{(\ell,m)}$ denote the output embedding of node i for modality m at the ℓ -th layer of the EDG module.

For a neighbor node j and modality m . The effective cross-modal signal $\mathbf{e}_j^{(\ell,m)}$ is computed as:

$$\mathbf{e}_j^{(\ell,m)} = \sum_{k=1}^K G\left(\mathbf{n}_j^{(\ell,m)}\right)_k \cdot E_k\left(\mathbf{n}_j^{(\ell,m)}\right), \quad (1)$$

$$\text{where } \mathbf{n}_j^{(\ell,m)} = \parallel_{m' \neq m} \mathbf{h}_j^{(\ell-1,m')},$$

and $E_k(\cdot)$ represents the k -th expert network (implemented as an MLP). Each expert specializes in discerning and extracting specific patterns of effective mutual information from the neighbor node set. K denotes the total number of expert. $G(\cdot)_k$ is the gating score indicating the relevance of the k -th expert, computed via a softmax gating network:

$$G\left(\mathbf{n}_j^{(\ell,m)}\right)_k = \frac{\exp\left(\text{MLP}_g(\mathbf{n}_j^{(\ell,m)})_k\right)}{\sum_{k'=1}^K \exp\left(\text{MLP}_g(\mathbf{n}_j^{(\ell,m)})_{k'}\right)}, \quad (2)$$

This ensures that the model dynamically selects experts to capture the diverse semantic combinations of neighboring modalities.

Topology-Aware Attention Mechanism. After mutual semantic information extraction, a Graph Transformer (Dwivedi & Bresson, 2020) layer is used to perform topology-aware attention mechanism:

$$\mathbf{h}_i^{(\ell,m)} = \text{GT}_\ell\left(q = \mathbf{h}_i^{(\ell-1,m)}; \mathcal{K}, \mathcal{V} = \{\mathbf{e}_j^{(\ell,m)}\}_{j \in \mathcal{N}_i}\right), \quad (3)$$

where $\mathcal{N}_i$ denotes the set of incoming neighbors for node i (including node i itself), $\text{GT}_\ell(\cdot)$ represents the ℓ -th layer of standard Graph Transformer. Crucially, distinct from standard cross-modal attention which typically operates within isolated instances (Radford et al., 2021), our mechanism explicitly integrates graph structural information into the cross-modal interaction process by querying the complementary modalities of neighboring nodes. See Appendix C.2 for the detailed formulations of our attention mechanism.

3.3. NDR for Modality Alignment

While the EDG module significantly enriches semantic representation by extracting and infusing cross-modal context, we argue that a robust MGFM requires a more explicit constraint to bridge the inherent semantic gap between modalities. We formulate modality alignment here as a global semantic consensus problem, to address this challenge, we introduce the NDR module at the node granularity.

Discretization Retrieval. We define Discretized Semantic Representation Space (DSRS) as $\mathcal{S} = \{s_1, \dots, s_C\}$ containing C learnable latent vectors, referred to as tokens. Let $\mathbf{h}_i^{(L,m)}$ denote the final output representation of node i for modality m from the EDG module. Each node embedding is mapped to its nearest DSRS token:

$$\mathbf{h}_i^{(cross,m)} = s_c, \text{ where } c = \underset{j}{\operatorname{argmin}} \|\mathbf{s}_j - \mathbf{h}_i^{(L,m)}\|_2, \quad (4)$$

where $\|\cdot\|_2$ denotes the Euclidean distance. This retrieval step forces features from different modalities to cluster around shared semantic space.

Text-Anchored General Knowledge Alignment. Inspired by Liu et al. (2025a), we propose a *General Knowledge Loss* that uses the *Text* modality (t) as an anchor. We align every other modality m to the text modality in the DSRS:

$$\mathcal{L}_{gen} = -\frac{1}{N(|\Omega|-1)} \sum_{m \neq t} \sum_{i=1}^N \left(\log \frac{z_{i,i}^{(t,m)}}{\sum_{j=1}^N z_{i,j}^{(t,m)}} + \log \frac{z_{i,i}^{(t,m)}}{\sum_{j=1}^N z_{j,i}^{(t,m)}} \right),$$

where $z_{i,j}^{(t,m)} = \exp \left(\operatorname{sim}(\mathbf{h}_i^{(cross,t)}, \mathbf{h}_j^{(cross,m)}) / \tau \right),$ (5)

and τ is a temperature parameter. This efficiently pulls the quantized representations of the same node across modalities together while pushing distinct nodes apart.

VQ Objective. To align the latent distribution of modality features with the quantized semantic space of DSRS, we utilize the loss function (Van Den Oord et al., 2017):

$$\mathcal{L}_{VQ} = \frac{1}{N} \sum_{i=1}^N \sum_{m \in \mathcal{M}} \left(\|sg[\mathbf{h}_i^{(cross,m)}] - \mathbf{h}_i^{(L,m)}\|_2^2 + \gamma \|\mathbf{h}_i^{(cross,m)} - sg[\mathbf{h}_i^{(L,m)}]\|_2^2 \right), \quad (6)$$

where $sg[\cdot]$ denotes the stop-gradient operator to handle the non-differentiable quantization, and γ controls the commitment cost.

3.4. Self-Supervised Training

Feature Reconstruction. To ensure robust semantic understanding, we propose a dual reconstruction mechanism. We minimize the error for recovering the masked input itself ($\mathcal{L}_{self}$) to capture modality-specific semantics, while

simultaneously enforcing cross-modal correlations by reconstructing features of complementary modalities:

$$\mathcal{L}_s = \frac{1}{|\Omega|N} \sum_{i=1}^N \sum_{m \in \mathcal{M}} \|\mathcal{D}_m^{SMR}(\mathbf{h}_i^{(all,m)}) - \mathbf{x}_i^{(m)}\|_2^2, \quad (7)$$

$$\mathcal{L}_c = \frac{\sum_{i=1}^N \sum_{m \neq m'} \|\mathcal{D}_{mm'}^{CMR}(\mathbf{h}_i^{(all,m)}) - \mathbf{x}_i^{(m')}\|_2^2}{(|\Omega|^2 - |\Omega|)N}, \quad (8)$$

where $\mathbf{x}_i^{(m)}$ is the original unmasked feature vector output by the modality encoder, $\mathcal{D}_m^{SMR}$ is the self modality reconstruction decoder (MLP), and $\mathcal{D}_{mm'}^{CMR}$ is the cross modality reconstruction decoder, both of which are implemented as MLPs. The total feature reconstruction loss is: $\mathcal{L}_{feat} = \mathcal{L}_s + \beta_{inter}\mathcal{L}_c$.

Structural Reconstruction. To preserve topological information, we employ a link reconstruction task. We use $\mathcal{L}_{topo}$ to denote the loss function, which enforces higher similarity scores for connected edges compared to randomly sampled negative pairs (details in Appendix C.3).

Total Objective. The final training objective combines all losses:

$$\mathcal{L} = \beta_1 \mathcal{L}_{feat} + \beta_2 \mathcal{L}_{topo} + \beta_3 \mathcal{L}_{gen} + \beta_4 \mathcal{L}_{VQ} + \beta_5 \mathcal{L}_{load}, \quad (9)$$

where β terms are hyperparameters balancing the contribution of each component, $\mathcal{L}_{load}$ is load balancing loss introduced in Appendix C.3.

3.5. Theoretical Analysis

3.5.1. WHY EDG CAPTURES SYNERGISTIC SEMANTICS?

Definition 3.1. Synergistic Features. Let $G^{(A)}$ and $G^{(B)}$ denote input graphs for modality A and B . It can be decomposed into independent modality-specific unique features $\{U_A, U_B\}$ and synergistic features $\{S_A, S_B\}$. $\{S_A, S_B\}$ satisfies the condition of zero mutual information under independent views, i.e., $I(Y; S_A) \approx 0$ and $I(Y; S_B) \approx 0$, but positive interaction information under a joint view, i.e., $I(Y; S_A, S_B) > 0$, where Y represents the latent semantic information related to the data.

Theorem 3.2. Synergy Preservation via EDG. Let $Z_{Vanilla}^*$ and Z_{EDG}^* denote the optimal representations learned by a vanilla Multimodal Graph Encoder (e.g., MMGCN) and PLANET with the EDG module, respectively. Under the compression constraint of the Information Bottleneck (Federici et al., 2020; Wu et al., 2020), provided that the trade-off parameter β is sufficiently small, the information gap between the two representations satisfies:

$$I(Y; Z_{EDG}^*) - I(Y; Z_{Vanilla}^*) \geq I(Y; S_A, S_B | U_A, U_B) > 0. \quad (10)$$

Such a theorem demonstrates that vanilla encoders inevitably discard synergistic features $\{S_A, S_B\}$ as noise. In contrast, EDG explicitly captures them, yielding a strictly larger information gain (proofs are shown in Appendix D).

Table 1. Main results on node classification tasks. We report Accuracy and F1-Macro for each dataset. Best results are highlighted in **bold**, the second-best are marked with underline. All baselines are first pre-trained and then fine-tuned.

Method	RedditS		Movies		Grocery		Toys		Ele-fashion		Goodreads-NC	
	Acc	F1-Macro	Acc	F1-Macro	Acc	F1-Macro	Acc	F1-Macro	Acc	F1-Macro	Acc	F1-Macro
GCN	92.44±0.62	87.34±1.22	52.26±0.74	42.35±1.23	78.60±0.41	64.90±1.09	77.63±0.72	75.54±0.97	85.33±0.06	68.01±0.23	78.44±0.06	68.12±0.16
MMGCN	90.27±0.34	84.22±0.73	53.41±1.03	41.66±2.03	82.56±0.49	73.83±0.93	<u>80.02±0.64</u>	76.36±1.23	86.59±0.08	68.85±0.35	<u>83.22±0.10</u>	71.28±0.22
MGAT	92.78±0.50	87.27±0.53	<u>53.87±0.50</u>	<u>44.09±1.60</u>	<u>83.74±0.62</u>	74.77±1.11	79.61±0.74	<u>77.09±0.87</u>	84.84±0.08	69.62±0.21	82.91±0.04	<u>71.45±0.11</u>
GRACE	93.01±0.53	<u>88.39±1.12</u>	48.09±0.97	37.18±1.33	70.83±0.81	60.69±1.05	72.82±0.66	69.09±0.63	83.58±0.11	70.09±0.47	74.96±0.06	70.09±0.11
GraphMAE2	92.81±0.44	87.93±0.37	50.08±0.77	38.68±1.63	76.24±0.60	66.74±1.33	75.11±0.52	71.80±0.50	83.32±0.31	65.92±0.59	74.15±0.22	69.20±0.28
RiemannGFM	91.63±0.45	85.20±1.13	52.80±0.43	40.74±1.25	82.62±0.46	<u>74.90±1.55</u>	77.85±0.45	74.84±0.60	87.07±0.20	<u>70.45±1.32</u>	78.13±0.17	70.73±0.24
GFT	93.02±0.37	87.00±2.03	51.33±0.67	28.14±1.82	76.80±2.22	59.11±3.02	79.52±0.58	76.00±0.92	<u>87.14±0.22</u>	70.33±1.24	75.93±0.41	66.18±0.30
SAMGPT	93.11±0.19	87.12±0.64	50.25±0.28	34.21±1.37	76.41±0.54	63.40±0.88	73.81±0.41	67.12±0.73	83.81±0.11	69.73±0.39	74.29±0.11	66.57±0.29
UniGraph2	<u>93.65±0.17</u>	87.91±0.68	53.02±0.53	43.43±1.86	82.10±0.37	73.93±1.37	79.00±0.59	76.02±0.78	87.06±0.18	69.80±0.82	79.06±0.27	68.74±0.36
PLANET	96.62±0.22	92.44±0.43	57.06±0.61	47.49±1.23	85.16±0.88	77.23±0.86	81.22±0.50	77.55±0.75	87.37±0.12	70.74±0.78	84.16±0.07	74.43±0.18
Rel-Improv.	↑ 3.17%	↑ 4.58%	↑ 5.92%	↑ 7.71%	↑ 1.70%	↑ 3.11%	↑ 1.50%	↑ 0.60%	↑ 0.26%	↑ 0.41%	↑ 1.13%	↑ 4.17%

Table 2. Link prediction and few-shot link classification results. We report MRR for link prediction, accuracy for few-shot link classification tasks.

Method	Link Prediction			Few-shot Link Classification						
	Amazon -Sports	Amazon -Cloth	Goodreads -LP	Amazon-Sports-2Way			Amazon-Cloth-2Way			
				10-shot	5-shot	3-shot	10-shot	5-shot	3-shot	
MMGCN	23.44±0.43	17.74±0.38	20.73±0.48	56.66±1.60	53.70±1.20	55.86±2.96	67.27±4.20	68.61±5.38	64.36±4.07	
MGAT	21.74±0.96	15.47±0.32	21.82±0.53	57.92±1.80	56.55±2.69	54.92±2.40	68.66±2.94	70.34±3.36	67.05±1.45	
GRACE	25.31±0.16	18.27±0.15	19.30±0.27	58.50±1.43	57.94±2.58	56.42±1.36	64.96±1.86	64.94±1.09	62.36±2.75	
GraphMAE2	24.54±0.30	18.69±0.21	19.99±0.19	56.05±1.70	54.36±1.65	53.28±3.09	62.33±1.78	61.39±1.71	60.95±1.61	
GFM	RiemannGFM	21.92±0.51	19.20±0.44	22.03±0.67	53.68±1.24	53.73±2.03	54.07±1.46	66.20±5.94	62.66±1.75	63.45±4.07
	GFT	22.04±0.62	17.63±0.59	20.16±1.21	55.73±3.36	56.86±2.30	56.75±2.63	65.37±4.03	66.66±2.10	63.70±2.84
	SAMGPT	24.09±0.22	16.41±0.20	24.91±0.38	60.48±4.25	59.00±3.56	59.41±3.62	74.25±1.75	74.20±1.60	72.61±1.90
MGFM	UniGraph2	27.09±0.13	19.31±0.35	19.44±0.19	65.08±3.17	64.27±2.74	60.83±3.90	73.77±1.91	71.28±4.20	73.44±1.57
	PLANET	27.51±0.14	20.25±0.27	27.62±0.25	67.84±1.38	64.36±3.05	62.89±3.99	75.44±2.84	75.22±1.20	74.03±4.12
Rel-Improv.	↑ 1.55%	↑ 4.87%	↑ 10.87%	↑ 4.24%	↑ 0.14%	↑ 3.39%	↑ 1.60%	↑ 1.37%	↑ 0.80%	

3.5.2. HOW NDR ENHANCES ALIGNMENT EFFICIENCY?

Definition 3.3. Push-forward Measure. Let $\hat{\mu}_m = \frac{1}{N} \sum_{i=1}^N \delta_{x_i^{(m)}}$ be the empirical measure of modality m , where δ is the Dirac measure, N is the number of samples. Given a DSRS $\mathcal{S}$ and a quantization function $Q(x) = \arg \min_{e \in \mathcal{S}} \|x - e\|_2$, we define the **Push-forward Measure** of modality m as $\hat{\nu}_m = Q_{\#} \hat{\mu}_m = \frac{1}{N} \sum_{i=1}^N \delta_{Q(x_i^{(m)})}$.

Theorem 3.4. Efficient Alignment via NDR. Assume the feature space is bounded. The alignment error between the empirical distribution of modality m ($\hat{\mu}_m$) and the anchor text modality t ($\hat{\mu}_t$) is bounded by:

$$W_1(\hat{\mu}_m, \hat{\mu}_t) \leq \mathbb{E}_{x \sim \hat{\mu}_m} \|x - Q(x)\|_2 + \mathbb{E}_{z \sim \hat{\mu}_t} \|z - Q(z)\|_2 + W_1(\nu_m^*, \nu_t^*) + O\left(\frac{C}{\sqrt{N}}\right), \quad (11)$$

where $W_1(\cdot, \cdot)$ denotes the 1-Wasserstein distance between distributions, C represents the DSRS size, and $W_1(\nu_m^*, \nu_t^*)$ represents the intrinsic bias between modalities.

This proves that projecting features into DSRS accelerates the alignment convergence rate from $O(N^{-1/d})$ in continuous spaces to $O(N^{-1/2})$. Additionally, minimizing $\mathcal{L}_{VQ}$

reduces the quantization error terms in Eq. (11), explicitly tightening the upper bound to ensure robust alignment.

4. Experiments

To validate the superiority of PLANET, we raise several questions: **Q1:** Does PLANET consistently outperform SOTA baselines across standard graph-centric tasks (i.e., node classification and link prediction) under both supervised learning and few-shot learning scenarios? **Q2:** How do **EDG** and **NDR** modules contribute to achieving topology-aware modality interaction and robust modality alignment, respectively? **Q3:** Can PLANET effectively support downstream generative tasks, thereby demonstrating the robust generalization capabilities and versatile applicability of a foundation model? **Q4:** How does PLANET fare in terms of computational efficiency compared to existing GFMs? The implementation details are introduced in Appendix E.

4.1. Graph-Centric Tasks

Experimental Settings. To answer **Q1**, we conduct extensive evaluations on node classification and link prediction

Table 3. Few-shot node classification results. We report accuracy for node classification tasks.

Method	Grocery-5way			Ele-fashion-5way			Goodreads-NC-5way		
	10-shot	5-shot	3-shot	10-shot	5-shot	3-shot	10-shot	5-shot	3-shot
MMGCN	53.73 $\pm$ 3.51	50.30 $\pm$ 3.13	48.13 $\pm$ 2.65	60.87 $\pm$ 3.08	57.05 $\pm$ 2.42	54.37 $\pm$ 2.99	56.42 $\pm$ 2.84	54.10 $\pm$ 2.90	52.93 $\pm$ 2.94
MGAT	55.53 $\pm$ 2.98	54.07 $\pm$ 3.18	50.83 $\pm$ 2.88	62.82 $\pm$ 2.18	61.10 $\pm$ 2.20	60.38 $\pm$ 2.57	58.22 $\pm$ 3.40	57.05 $\pm$ 2.35	53.15 $\pm$ 3.09
GRACE	62.22 $\pm$ 2.89	59.22 $\pm$ 3.50	57.28 $\pm$ 4.91	64.72 $\pm$ 3.04	59.90 $\pm$ 2.73	55.49 $\pm$ 4.10	59.20 $\pm$ 2.00	61.53 $\pm$ 3.14	56.84 $\pm$ 4.65
GraphMAE2	58.00 $\pm$ 4.26	54.60 $\pm$ 3.83	51.90 $\pm$ 5.76	63.65 $\pm$ 3.48	60.00 $\pm$ 3.53	57.35 $\pm$ 3.05	49.83 $\pm$ 2.60	46.30 $\pm$ 2.95	43.20 $\pm$ 2.70
GFM	RiemannGFM	67.14 $\pm$ 2.02	66.23 $\pm$ 2.73	63.63 $\pm$ 2.48	61.73 $\pm$ 2.90	60.24 $\pm$ 3.55	59.13 $\pm$ 3.71	60.54 $\pm$ 3.92	57.17 $\pm$ 3.83
	GFT	63.12 $\pm$ 4.07	64.60 $\pm$ 3.54	61.45 $\pm$ 2.22	62.28 $\pm$ 2.62	61.38 $\pm$ 3.41	59.08 $\pm$ 3.95	51.62 $\pm$ 4.73	50.95 $\pm$ 4.35
	SAMGPT	66.47 $\pm$ 10.67	62.33 $\pm$ 10.49	52.73 $\pm$ 12.69	61.27 $\pm$ 10.82	61.13 $\pm$ 9.32	54.20 $\pm$ 10.78	51.80 $\pm$ 8.65	47.00 $\pm$ 8.35
MGFM	UniGraph2	66.27 $\pm$ 3.11	61.25 $\pm$ 3.09	60.05 $\pm$ 4.13	60.38 $\pm$ 2.08	58.73 $\pm$ 2.95	53.98 $\pm$ 4.07	63.80 $\pm$ 4.02	61.55 $\pm$ 3.97
	PLANET	81.93 $\pm$ 3.48	79.88 $\pm$ 3.27	77.85 $\pm$ 4.06	74.85 $\pm$ 3.73	72.97 $\pm$ 3.55	70.50 $\pm$ 4.37	69.18 $\pm$ 3.90	67.59 $\pm$ 4.68
	Rel-Improv.	$\uparrow$ 22.03%	$\uparrow$ 20.61%	$\uparrow$ 22.35%	$\uparrow$ 15.65%	$\uparrow$ 18.88%	$\uparrow$ 16.76%	$\uparrow$ 8.43%	$\uparrow$ 9.81%
		$\uparrow$ 6.62%							

tasks under both supervised learning and few-shot learning scenarios. We compare PLANET with 9 strong baselines, which can be categorized into five distinct groups: (1) **Vanilla GNNs**: GCN (Kipf & Welling, 2017). (2) **Multimodal Graph Models**: Including MMGCN (Wei et al., 2019) and MGAT (Tao et al., 2020). (3) **Self-supervised Graph Learning Models**: Comprising contrastive learning method, GRACE (Zhu et al., 2020) and generative learning method, GraphMAE2 (Hou et al., 2023). (4) **Graph Foundation Models**: Including SAMGPT (Yu et al., 2025) and RiemannGFM (Sun et al., 2025), which focus on learning unified graph structures, and GFT (Wang et al., 2024b), which targets TAGs. (5) **Multimodal Graph Foundation Models**: Specifically UniGraph2 (He et al., 2025b), a novel baseline explicitly designed for processing MAGs. Please refer to Appendix E for detailed implementation of baselines.

Supervised Learning. Table 1 and 2 show the results of node classification and link prediction in supervised learning. The results indicate that PLANET consistently outperforms baselines across all datasets. We attribute this superior performance to the synergistic collaboration between the EDG and NDR modules operating at distinct granularities. By jointly facilitating efficient topology-aware modality interaction and alignment, PLANET ensures the generation of high-quality node embeddings.

Few-Shot Learning. Tables 2 and 3 present the results for few-shot node classification and link prediction, respectively. In these challenging low-resource scenarios, PLANET continues to outperform baselines across all datasets. Most notably, in few-shot node classification tasks, PLANET achieves remarkable performance gains (15.68%). These results underscore the model’s robust learning and generalization capabilities under conditions of severe data scarcity, providing evidence that our pre-training paradigm effectively enables the model to acquire comprehensive prior knowledge, which can be efficiently transferred to downstream tasks with minimal supervision.

Table 4. Ablation studies on PLANET key components.

	Toys	RedditS	Ele-fashion	Amazon-Sports
<i>EDG for Modality Interaction</i>				
w/o. MoE	77.77 $\pm$ 0.64	95.19 $\pm$ 0.30	85.33 $\pm$ 0.27	25.92 $\pm$ 0.16
w/o. MoE+AM	76.33 $\pm$ 0.54	93.01 $\pm$ 0.16	84.52 $\pm$ 0.14	24.76 $\pm$ 0.13
<i>NDR for Modality Alignment</i>				
w/o. DSRS	78.53 $\pm$ 0.80	95.12 $\pm$ 0.33	85.93 $\pm$ 0.18	26.19 $\pm$ 0.22
w/o. $\mathcal{L}_{gen}$	77.85 $\pm$ 0.73	94.97 $\pm$ 0.31	84.71 $\pm$ 0.22	25.38 $\pm$ 0.19
PLANET	81.22$\pm$0.50	96.62$\pm$0.22	87.37$\pm$0.12	27.51$\pm$0.14

4.2. Ablation Study

To address Q2, we conduct rigorous ablation studies removing core components of PLANET, as shown in Table 4.

❶ **w/o. MoE Component.** We remove the MoE mechanism. Here, neighboring multimodal features are fed directly into the attention mechanism. The result confirms that the MoE is essential for extracting effective cross-modal signals before modality interaction. ❷ **w/o. MoE + Attention Mechanism.** We investigate the impact of topology-aware modality interaction by replacing the entire EDG module with standard, independent Graph Transformers for each modality. The significant performance degradation validates the effectiveness of explicitly modeling the topology-aware interplay between modalities. ❸ **w/o. DSRS.** We remove the DSRS and the associated Discretization Retrieval, directly applying the General Knowledge Loss ($\mathcal{L}_{gen}$) to the continuous embeddings $\mathbf{h}_i^{(L,m)}$. The resulting performance drop confirms its critical role in modality alignment by projecting distinct modalities into a unified semantic space. ❹ **w/o. General Knowledge Loss.** We retain the DSRS structure but disable the text-anchored General Knowledge Loss ($\mathcal{L}_{gen}$). Results show that without this signal, the model fails to effectively bridge the semantic gap between the quantized tokens of different modalities, leading to suboptimal alignment.

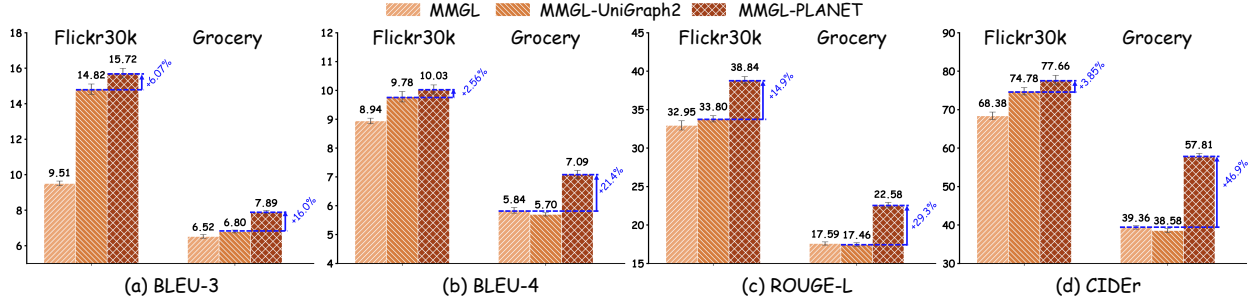


Figure 4. G2Text generation results. We report BLEU-3, BLEU-4, ROUGE-L, and CIDEr on the Flickr30k and Grocery datasets.

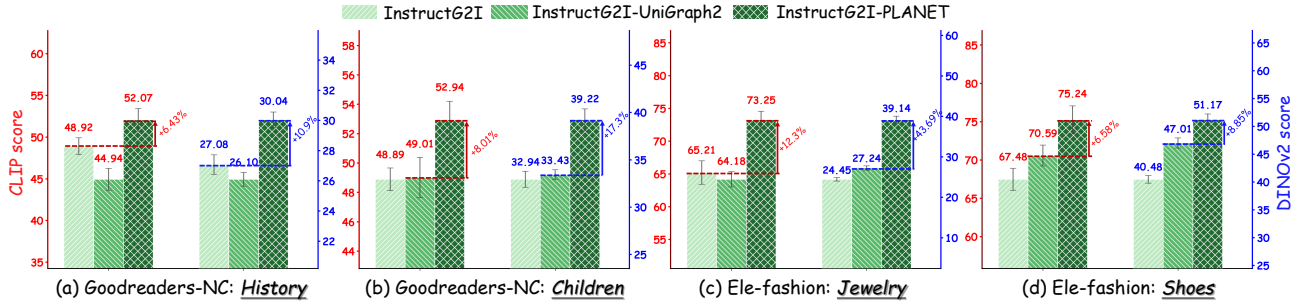


Figure 5. G2Image generation results. We report CLIP scores in red and DINOv2 scores in blue across four categories selected from the Goodreads-NC and Ele-fashion datasets.

4.3. Multimodal Generative Tasks

Experimental Settings. Leveraging the rich cross-domain semantic information and universal topological patterns learned during pre-training, PLANET exhibits strong potential for generative applications. To address Q3, we conduct evaluations on two distinct tasks: ❶ **Graph-to-Text (G2Text) Generation.** This task aims to generate a comprehensive textual description for a target node, conditioned on the graph structure and multimodal context. We strictly follow the evaluation settings of MMGL (Yoon et al., 2023). Specifically, MMGL relies on frozen CLIP encoders, and we replace these embeddings with the text and image embeddings generated by PLANET. ❷ **Graph-to-Image (G2Image) Generation.** This task focuses on synthesizing an image for a target node based on a text prompt and neighbor image context. Adopting the framework of InstructG2I (Jin et al., 2024), we replace the image features extracted by the pre-trained CLIP image encoder with the topology-enriched image embeddings produced by PLANET. Implementation details are provided in Appendix E.3.

Results. Fig. 4 and 5 shows the results of G2Text and G2Image tasks, respectively. PLANET consistently outperforms baselines across all metrics. We attribute this to our Divide-and-Conquer strategy, which produces superior embeddings through modality interaction and alignment.

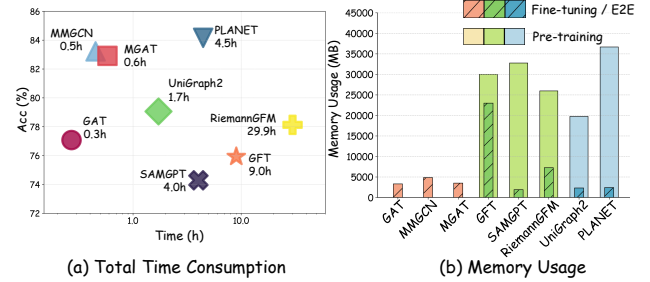


Figure 6. Efficiency comparison between models on Goodreads-NC. (a) Total time consumption. (b) GPU memory usage during different stages.

4.4. Computational Efficiency Analysis

To answer Q4, we conducted an efficiency analysis on the large-scale Goodreads-NC dataset. We evaluated the total pre-training and fine-tuning time of various baselines. For task-specific models (i.e., GAT, MMGCN, MGAT), we record their End-to-End (E2E) training time. Additionally, we monitor the GPU memory usage across different stages. Results (Fig. 6) demonstrate that PLANET achieves a superior efficiency-performance trade-off, validating the scalability of our Divide-and-Conquer design. PLANET utilizes sufficient memory for pre-training, its fine-tuning memory usage is minimal since only the linear classifier is trained. This indicates that our model enables resource-efficient adaptation for downstream tasks.

5. Conclusion

In this work, we propose PLANET to resolve the critical limitations of existing MGFMs in modality interaction and alignment. Leveraging a Divide-and-Conquer strategy, the EDG module couples expert-driven extraction with topology-aware attention to facilitate modality interaction at embedding level, and the NDR module bridges the semantic gap by mapping multimodal representations into a unified Discretized Semantic Representation Space at node level. Theoretical analysis and extensive experiments confirm the superiority of PLANET, which achieves state-of-the-art performance across diverse graph-centric and multimodal generative tasks, establishing a robust and scalable framework for learning on MAGs. In future work, we aim to extend PLANET to incorporate richer modalities like audio and video, broadening the applicability of MGFMs in web-scale scenarios.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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Table 5. Statistics of MAG datasets.

Domain	Dataset	Avg. #Nodes	Avg. #Edges	#Graphs	Task	#Classes	#Pretrain Weights
E-commerce Network	Movies	16,672	218,390	1	NC	20	5.0
	Toys	20,695	126,886	1	NC	18	5.0
	Grocery	17,074	171,340	1	NC	20	5.0
	Amazon-Sports	50,250	356,202	1	LP	–	1.0
	Amazon-Cloth	125,839	951,271	1	LP	–	1.0
	Ele-fashion	97,766	199,602	1	NC	12	1.0
Social Network	Reddit-S	15,894	566,160	1	NC	20	10.0
	Flickr30k	31,783	181,551	1	–	–	0.0
Knowledge Graph	MM-CoDEX-s	1,383	15,884	1	KGC	–	20.0
	MM-CoDEX-m	7,697	52,840	1	KGC	–	20.0
Books Network	Goodreads-NC	685,294	7,235,048	1	NC	11	0.5
	Goodreads-LP	636,502	3,437,017	1	LP	–	0.5

A. Datasets

Dataset. Detailed information of each dataset is presented in Table 5.

- **MAGB Datasets.** We select four representative graphs from the MAGB benchmark (Yan et al., 2025). **Movies**, **Toys** and **Grocery** are e-commerce networks constructed from Amazon. Nodes represent products, and edges indicate also-bought or also-viewed co-purchasing relationships. Each node is associated with a product description and a product image. **Reddit-S** is a social network graph derived from the Reddit platform. Nodes represent user posts containing both text and images. Edges connect posts commented on by the same user, reflecting shared user interests.
- **MM-GRAPH Datasets.** We incorporate seven datasets from the MM-GRAPH benchmark (Zhu et al., 2025a). **Amazon-Sports**, **Amazon-Cloth** and **Ele-fashion** are e-commerce networks where nodes represent items associated with product titles and images. **Goodreads-NC** and **Goodreads-LP** are books networks enriched with cover and descriptions. Edges represent user co-interactions, linking books read or liked by the same users. **MM-CoDEX-s** and **MM-CoDEX-m** are knowledge graphs where entities contain Wikipedia texts and images. Edges denote semantic relations (e.g., born_in) between entities.
- **Flickr30k.** Flickr30k is a social network. Notably, Flickr30k lacks an intrinsic graph structure. To adapt it for graph learning, we construct a k-NN graph, where edges are established between nodes with high feature similarity to capture latent semantic correlations. Nodes represent images associated with five descriptions.

Dataset Split. For the MAGB datasets and Flickr30k, each dataset is randomly partitioned into training sets (60%), validation sets (20%) and testing sets (20%). For the MM-GRAPH datasets, we strictly adhere to the official data splits provided in the original benchmark (Zhu et al., 2025a).

B. Detailed Implementation of Empirical Study

B.1. Detailed Implementation of Empirical Study 1

The first empirical study (Fig. 1(a)&(b)) is designed to validate the benefits of incorporating multimodal information and the superiority of MGFMs approaches. we conducted a comparative study involving both traditional MGL models (MMGCN, MGAT), Graph Foundation Models (RiemannGFM, SAMGPT) and Multimodal Graph Foundation Models (UniGraph2).

Input Modality Settings. We standardized the feature dimensions and encoding processes across all settings. Specifically, both text data and image data are encoded through Qwen2-VL-7B-Instruct (Wang et al., 2024a). The output features are in a fixed dimension of 3,584. For *Text* or *Image* settings, These settings represent single-modality scenarios. we adopted a duplication strategy to ensure compatibility with model architectures which need dual modality inputs. For instance, in the *Text* setting, we utilize the text embeddings as the primary input and duplicate them to serve as the input for the

second modality (i.e., replacing the image features with text features). For the *Text+Image* setting, we provide models with comprehensive multimodal information (i.e., text embeddings and image embeddings).

Model Adaptation. For models inherently designed to handle MAGs, including MMGCN, MGAT, and UniGraph2, we retain their original architectures without modification. For GFM that are not naturally designed to process MAGs, including RiemannGFM and SAMGPT, we concatenated the embeddings from the distinct modalities (e.g., for the *Text+Image* setting, we concatenated the text and image embeddings) to form a unified node feature embedding.

Results Analysis. As shown in Fig. 1(a)&(b), the *Text+Image* setting consistently outperforms single-modality baselines, corresponding to the conclusion: **① The use of multiple modalities consistently improve model performance.** Comparing UniGraph2 with RiemannGFM and SAMGPT, UniGraph2 significantly outperforms general GFMs on both dataset across all settings, corresponding to the conclusion: **② MGFM significantly outperform GFMs.**

B.2. Detailed Implementation of Empirical Study 2

This empirical study (Fig. 1(c)&(d)) is designed to validate the limitations of recent MGFM, exemplified by the state-of-the-art UniGraph2.

Modality Interaction (MI). The original UniGraph2 employs an “early fusion” strategy, where embeddings from different modalities are aggregated via weighted sum before entering graph encoder. To realize MI, we removed the early fusion mechanism and introduced a naive layer-wise interaction strategy. Specifically, in each Graph Encoder layer, we allow information to flow from one modality m' to another modality m via a linear projection:

$$\mathbf{h}^{(\ell,m)} = \text{GNNLayer}_m^{(l)} \left(\mathbf{h}^{(\ell-1,m)} \right) + \frac{\alpha}{|\Omega| - 1} \sum_{m' \neq m} \mathbf{W}_{m'm}^{(l)} \cdot \text{GNNLayer}_{m'}^{(l)} \left(\mathbf{h}^{(\ell-1,m')} \right) \quad (12)$$

where $\mathbf{W}_{m'm}^{(l)}$ is a learnable projection matrix in layer l , capturing the relationship between modalities m' and m . α is a hyperparameter.

MI+Modality Alignment (MA). Building upon the MI, we incorporated the Symmetric InfoNCE Loss (Radford et al., 2021) to enforce explicit modality alignment. This contrastive objective pulls representations of the same node across different modalities closer in the latent space.

C. Detailed Formulations

C.1. Modality Masking

Inspired by Hou et al. (2022), we employ a modality masking strategy. Formally, for each node $v_i \in \tilde{\mathcal{V}}$ and modality $m \in \Omega$, we have:

$$\tilde{\mathbf{x}}_i^{(m)} = \mathbf{x}_i^{(m)} \odot \mathbf{b}_i^{(m)}, \quad (13)$$

where $\mathbf{b}_i^{(m)} \in \{0, 1\}^{d_m}$ is a binary mask vector sampled from a Bernoulli distribution, $\tilde{\mathcal{V}} \subset \mathcal{V}$ is a sampled subset of nodes, and $\odot$ denotes the element-wise product. This process encourages the model to recover missing feature dimensions using contextual information from the graph and other modalities.

C.2. Topology-Aware Attention Mechanism

In Sec. 3.2, we abstract the topology-aware attention mechanism as a generic operator $\text{GT}_l(\cdot)$. Here, we provide its detailed mathematical formulation. We strictly follow the implementation of Graph Transformer (Dwivedi & Bresson, 2020), but change the *query*, *key* and *value* vectors. The output of the multi-head attention is computed as:

$$\hat{\mathbf{h}}_i^{(\ell,m)} = \mathbf{O}^{(\ell,m)} \parallel_{k=1}^H \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij,k}^{(\ell,m)} \mathbf{V}_k^{(\ell,m)} \mathbf{e}_j^{(\ell,m)} \right), \quad (14)$$

where H represents the number of heads. $\parallel$ denotes the concatenation operation, $\mathcal{N}_i$ denotes the set of incoming neighbors for node i , and $\mathbf{O}^{(\ell,m)}$ is the output projection matrix. For the k -th head, $\mathbf{V}_k^{(\ell,m)}$ maps the expert-distilled neighbor signal $\mathbf{e}_j^{(\ell,m)}$ to the *value* vector. The attention coefficient $\alpha_{ij,k}^{(\ell,m)}$ determines the importance of neighbor j 's cross-modal

information to node i :

$$\alpha_{ij,k}^{(\ell,m)} = \text{softmax}_j \left(\frac{\left(\mathbf{Q}_k^{(\ell,m)} \mathbf{h}_i^{(\ell-1,m)} \right)^\top \left(\mathbf{K}_k^{(\ell,m)} \mathbf{e}_j^{(\ell,m)} \right)}{\sqrt{d_k}} \right), \quad (15)$$

where d_k is the dimension of each head. The projection matrices $\mathbf{Q}_k^{(\ell,m)}, \mathbf{K}_k^{(\ell,m)}$ transform the input features into *query* and *key* vectors, respectively. After that, a residual connection followed by Layer Normalization (Ba et al., 2016) is applied to the output of the attention mechanism:

$$\hat{\mathbf{h}}_i^{(\ell,m)} = \text{LayerNorm} \left(\mathbf{h}_i^{(\ell-1,m)} + \hat{\mathbf{h}}_i^{(\ell,m)} \right). \quad (16)$$

The normalized representation is then passed through a Feed-Forward Network (FNN):

$$\hat{\mathbf{h}}_i^{(\ell,m)} = \text{MLP}_\ell^2 \left(\text{ReLU} \left(\text{MLP}_\ell^1 \left(\hat{\mathbf{h}}_i^{(\ell,m)} \right) \right) \right). \quad (17)$$

Finally, a second residual connection and Layer Normalization are applied to produce the final node embedding for layer l :

$$\mathbf{h}_i^{(\ell,m)} = \text{LayerNorm} \left(\hat{\mathbf{h}}_i^{(\ell,m)} + \hat{\mathbf{h}}_i^{(\ell,m)} \right). \quad (18)$$

C.3. Loss Functions

MoE Load Balancing. To prevent the issue of expert collapse (i.e., a small subset of experts dominates the processing while others remain idle), we consider the MoE load balancing loss. Adopting the standard formulation proposed by Fedus et al. (2022):

$$\mathcal{L}_{load} = K \cdot \sum_{k=1}^K P_k f_k, \quad (19)$$

where f_k is the fraction of samples assigned to expert k , and P_k denotes the average routing probability for expert k across a batch.

Structural Reconstruction. To preserve topological information, we minimize the binary cross-entropy loss over observed edges $\mathcal{E}$ (positive samples) and randomly sampled unconnected pairs $\hat{\mathcal{E}}$ (negative samples):

$$\mathcal{L}_{topo} = \sum_{m \in \Omega} \left[-\frac{1}{|\Omega||\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \log \left(\sigma \left(\mathbf{u}_i^{(m)\top} \cdot \mathbf{u}_j^{(m)} \right) \right) - \frac{1}{|\Omega||\hat{\mathcal{E}}|} \sum_{(i',j') \in \hat{\mathcal{E}}} \log \left(1 - \sigma \left(\mathbf{u}_{i'}^{(m)\top} \cdot \mathbf{u}_{j'}^{(m)} \right) \right) \right], \quad (20)$$

where SRD is a structural projection head, σ is the sigmoid function, and $\mathbf{u}_i^{(m)} = \mathcal{D}_m^{SR}(\mathbf{h}_i^{(all,m)})$, $\mathcal{D}_m^{SR}$ is structural reconstruction decoder for modality m , which is an MLP.

D. Proof of Theoretical Analysis

D.1. Proof of Theorem 3.2

In this section, we utilize the Information Bottleneck (IB) principle to prove that vanilla Multimodal Graph Encoders (e.g., MMGCN) inevitably discard synergistic features, while PLANET’s topology-aware interaction preserves them.

Following the formulation in Graph Information Bottleneck (Wu et al., 2020), the goal of graph representation learning is to maximize the IB Lagrangian $\mathcal{L}_{IB}(Z) = I(Y; Z) - \beta I(X; Z)$, where $\beta > 0$ controls the compression trade-off.

Vanilla Multimodal Graph Encoder. The Vanilla Multimodal Graph Encoder processes modalities independently. For a specific modality A , the encoding follows the Markov chain $Y \leftrightarrow G^{(A)} \rightarrow Z_A$. The optimization objective is to maximize the local IB: $\mathcal{L}_{Vanilla} = I(Y; Z_A) - \beta I(G^{(A)}; Z_A)$. To prove that this encoder discards synergistic features, We define two potential encoding strategies:

- **Drop Synergistic Features (DSF):** The encoder captures only unique features, i.e., $Z_A^- \approx \{U_A\}$.

- **Keep Synergistic Features (KSF):** The encoder captures both unique and synergistic features, i.e., $Z_A^+ \approx \{U_A, S_A\}$.

We calculate the Marginal Contribution $\Delta\mathcal{L}_{Vanilla}$ of shifting from Strategy DSF to Strategy KSF, which can be expressed as $\Delta\mathcal{L}_{Vanilla} = \Delta I_Y - \beta \cdot \Delta I_{G^{(A)}}$. For ΔI_Y , we apply the Chain rule of mutual information (Federici et al., 2020):

$$\begin{aligned}\Delta I_Y &= I(Y; Z_A^+) - I(Y; Z_A^-) \\ &= I(Y; U_A, S_A) - I(Y; U_A) \\ &= I(Y; U_A) + I(Y; S_A | U_A) - I(Y; U_A) \\ &= I(Y; S_A | U_A).\end{aligned}\tag{21}$$

Crucially, under the independent encoding setting, modality B is unobservable. According to Definition 3.1, the synergistic feature S_A contains no information about Y without the presence of S_B (i.e., $I(Y; S_A | U_A) \approx 0$). Therefore, we have $\Delta I_Y \approx 0$.

For $\Delta I_{G^{(A)}}$, we similarly apply the Chain rule:

$$\begin{aligned}\Delta I_{G^{(A)}} &= I(G^{(A)}; Z_A^+) - I(G^{(A)}; Z_A^-) \\ &= I(G^{(A)}; U_A, S_A) - I(G^{(A)}; U_A) \\ &= I(G^{(A)}; S_A | U_A) \\ &= H(S_A | U_A) - H(S_A | G^{(A)}, U_A).\end{aligned}\tag{22}$$

Since S_A is intrinsically part of the input $G^{(A)}$, we have $H(S_A | G^{(A)}, U_A) = 0$, therefore:

$$\Delta I_{G^{(A)}} = H(S_A | U_A) > 0.\tag{23}$$

Substituting these results back to the Marginal Contribution $\Delta\mathcal{L}_{Vanilla}$:

$$\Delta\mathcal{L}_{Vanilla} \approx 0 - \beta \cdot H(S_A | U_A) < 0.\tag{24}$$

Since the contribution is negative for any $\beta > 0$, maximizing the local IB objective necessitates the exclusion of synergistic features to avoid incurring unnecessary compression costs. Consequently, we have $Z_{Vanilla}^* \approx \{U_A, U_B\}$.

PLANET with the EDG Module. The topology-aware interaction mechanism aggregates semantic context from the cross-modal neighborhood. This implies that the encoding process follows a unified Markov chain $Y \leftrightarrow \{G^{(A)}, G^{(B)}\} \rightarrow Z_{EDG}$. The optimization objective is to maximize the local IB: $\mathcal{L}_{EDG} = I(Y; Z_{EDG}) - \beta I(G^{(A)}, G^{(B)}; Z_{EDG})$.

Similarly, we contrast two strategies: The first strategy captures only unique features, i.e., $Z_{EDG}^- = \{U_A, U_B\}$, and the second strategy additionally captures the synergistic pair, i.e., $Z_{EDG}^+ = \{U_A, U_B, S_A, S_B\}$. The marginal contribution of transitioning from the first strategy to the second strategy is formulated as:

$$\Delta\mathcal{L}_{EDG} = \Delta I_Y - \beta \cdot \Delta I_{\{G^{(A)}, G^{(B)}\}}.\tag{25}$$

Applying the chain rule yields $\Delta I_{\{G^{(A)}, G^{(B)}\}} = H(S_A, S_B | U_A, U_B) > 0$ and $\Delta I_Y = I(Y; S_A, S_B | U_A, U_B)$. Under the joint view enabled by the unified Markov chain, the synergistic features become informative. By Definition 3.1, the interaction information is strictly positive, yielding:

$$\Delta I_Y = I(Y; S_A, S_B | U_A, U_B) \gg 0.\tag{26}$$

Consequently, we have $\Delta\mathcal{L}_{EDG} = \Delta I_Y - \beta \cdot H(S_A, S_B | U_A, U_B)$. Given that the significant synergistic information gain outweighs the weighted compression cost (i.e., $\Delta I_Y > \beta \cdot \Delta I_{\{G^{(A)}, G^{(B)}\}}$), $\Delta\mathcal{L}_{EDG}$ becomes strictly positive. Consequently, maximizing the joint IB objective necessitates the preservation of these features, leading to $Z_{EDG}^* \approx \{U_A, U_B, S_A, S_B\}$.

Finally, we quantify the information gap between the two representations:

$$I(Y; Z_{EDG}^*) - I(Y; Z_{Vanilla}^*) \approx I(Y; U_A, U_B, S_A, S_B) - I(Y; U_A, U_B) = I(Y; S_A, S_B | U_A, U_B) > 0.\tag{27}$$

This theorem demonstrates that our EDG module captures a strictly larger amount of relevant information than vanilla Multimodal Graph Encoders which process MAGs as independent graphs.

D.2. Proof of Theorem 3.4

To analyze the generalization bound of our alignment strategy, we employ the 1-Wasserstein distance $W_1(\hat{\mu}_m, \hat{\mu}_t)$ as the metric for distributional discrepancy.

Utilizing the triangle inequality property of the Wasserstein metric, we decompose the total alignment error into three components:

$$W_1(\hat{\mu}_m, \hat{\mu}_t) \leq W_1(\hat{\mu}_m, \hat{\nu}_m) + W_1(\hat{\nu}_m, \hat{\nu}_t) + W_1(\hat{\nu}_t, \hat{\mu}_t). \quad (28)$$

For $W_1(\hat{\mu}_m, \hat{\nu}_m)$. By the Kantorovich formulation of Optimal Transport (Villani et al., 2008), the Wasserstein distance is defined as the infimum of transport costs over all valid joint couplings $\Pi(\hat{\mu}_m, \hat{\nu}_m)$:

$$W_1(\hat{\mu}_m, \hat{\nu}_m) = \inf_{\pi \in \Pi(\hat{\mu}_m, \hat{\nu}_m)} \mathbb{E}_{(x,y) \sim \pi} [\|x - y\|_2]. \quad (29)$$

We define π^* such that it transports the probability mass $1/N$ associated with each sample $x_i^{(m)}$ directly to its corresponding quantized token $Q(x_i^{(m)})$. Formally, the joint distribution π^* is supported exclusively on the set of pairs $\{(x_i^{(m)}, Q(x_i^{(m)}))\}_{i=1}^N$. Let $\mathcal{C}(\pi)$ denote the transport cost associated with a coupling π . Under our deterministic coupling π^* , this cost is exactly:

$$\mathcal{C}(\pi^*) = \frac{1}{N} \sum_{i=1}^N \|x_i^{(m)} - Q(x_i^{(m)})\|_2 = \mathbb{E}_{x \sim \hat{\mu}_m} \|x - Q(x)\|_2. \quad (30)$$

Therefore, we have the inequality:

$$W_1(\hat{\mu}_m, \hat{\nu}_m) \leq \mathcal{C}(\pi^*) = \mathbb{E}_{x \sim \hat{\mu}_m} \|x - Q(x)\|_2. \quad (31)$$

For $W_1(\hat{\nu}_t, \hat{\mu}_t)$. Similarly, due to the symmetry of the Wasserstein metric and the same quantization mechanism applied to the anchor text modality t , we have the corresponding upper bound:

$$W_1(\hat{\nu}_t, \hat{\mu}_t) = W_1(\hat{\mu}_t, \hat{\nu}_t) \leq \mathbb{E}_{x \sim \hat{\mu}_t} \|x - Q(x)\|_2. \quad (32)$$

For $W_1(\hat{\nu}_m, \hat{\nu}_t)$. We denote ν_m^* and ν_t^* as the underlying ground-truth discrete distributions for modality m and the anchor text modality t , both supported on $\mathcal{S}$. Accordingly, the empirical push-forward measures $\hat{\nu}_m$ and $\hat{\nu}_t$ are viewed as samples drawn from these corresponding ground-truth distributions. Leveraging the triangle inequality, we have:

$$W_1(\hat{\nu}_m, \hat{\nu}_t) \leq W_1(\hat{\nu}_m, \nu_m^*) + W_1(\nu_m^*, \nu_t^*) + W_1(\nu_t^*, \hat{\nu}_t). \quad (33)$$

Let $\delta \in (0, 1)$ be the scale parameter. According to Proposition 1 in Weed & Bach (2019), The 1-Wasserstein distance is bounded by:

$$\mathbb{E}[W_1(\hat{\nu}_m, \nu_m^*)] \lesssim \delta^{k^*} + \sum_{k=1}^{k^*} \delta^{k-1} \sum_{Q_i^k \in \mathcal{Q}^k} \mathbb{E} [|\hat{\nu}_m(Q_i^k) - \nu_m^*(Q_i^k)|], \quad (34)$$

where k^* is the truncation depth, and $\mathcal{Q}^k$ is the partition at scale k .

We simplify the bound by observing three key properties: ① For a sufficiently large truncation depth k^* , the partition resolution exceeds the minimum separation of DSRS vectors, thereby eliminating the truncation error (i.e., $\delta^{k^*} \rightarrow 0$). ② The finite support restricts the inner summation $\sum_{Q_i^k \in \mathcal{Q}^k}$ to at most C non-zero terms. ③ The term $\mathbb{E}[|\hat{\nu}_m(Q_i^k) - \nu_m^*(Q_i^k)|]$ corresponds to the mean absolute deviation of empirical frequencies. By Jensen's inequality and the variance bound of the binomial distribution, this is strictly bounded by $\sqrt{\frac{1}{4N}} = \frac{1}{2\sqrt{N}}$.

Substituting these properties into Eq. (34):

$$\begin{aligned}
 \mathbb{E}[W_1(\hat{\nu}_m, \nu_m^*)] &\leq \sum_{k=1}^{\infty} \delta^{k-1} \sum_{Q_i^k \in \mathcal{Q}^k} \mathbb{E}[|\hat{\nu}_m(Q_i^k) - \nu_m^*(Q_i^k)|] \\
 &\leq \sum_{k=1}^{\infty} \delta^{k-1} \cdot \left(\sum_{j=1}^C \frac{1}{2\sqrt{N}} \right) \\
 &= \left(\sum_{k=1}^{\infty} \delta^{k-1} \right) \cdot \frac{C}{2\sqrt{N}}.
 \end{aligned} \tag{35}$$

Since $\delta \in (0, 1)$, the geometric series converges to a constant C_δ , yielding $\mathbb{E}[W_1(\hat{\nu}_m, \nu_m^*)] \leq C_\delta \cdot \frac{C}{\sqrt{N}} = O\left(\frac{C}{\sqrt{N}}\right)$. Both $W_1(\hat{\nu}_m, \nu_m^*)$ and $W_1(\nu_t^*, \hat{\nu}_t)$ satisfy this bound, the total discrete alignment error converges at the rate:

$$\mathbb{E}[W_1(\hat{\nu}_m, \hat{\nu}_t)] \leq \mathbb{E}[W_1(\hat{\nu}_m, \nu_m^*)] + \mathbb{E}[W_1(\nu_m^*, \nu_t^*)] + \mathbb{E}[W_1(\nu_t^*, \hat{\nu}_t)] \leq \mathbb{E}[W_1(\nu_m^*, \nu_t^*)] + O\left(\frac{C}{\sqrt{N}}\right) \tag{36}$$

Substituting Eq. (32), (36), (31) into Eq. (28), we obtain the final bound:

$$W_1(\hat{\mu}_m, \hat{\mu}_t) \leq \mathbb{E}_{x \sim \hat{\mu}_m} \|x - Q(x)\|_2 + \mathbb{E}_{z \sim \hat{\mu}_t} \|z - Q(z)\|_2 + W_1(\nu_m^*, \nu_t^*) + O\left(\frac{C}{\sqrt{N}}\right). \tag{37}$$

Comparison. Consider the standard case in continuous space $\mathbb{R}^d$. According to Theorem 1 and Proposition 7 in [Weed & Bach \(2019\)](#), the convergence rate of Wasserstein estimation is governed by the lower Wasserstein dimension $d_*(\mu)$ of the measure. For continuous feature distributions in high-dimensional spaces (e.g., dense semantic embeddings in $\mathbb{R}^d$), the intrinsic dimension $d_*(\mu)$ typically equals the ambient dimension d . Consequently, we have:

$$\mathbb{E}[W_1(\hat{\mu}, \mu)] \gtrsim N^{-1/d}. \tag{38}$$

Conclusion. Eq. (38) implies that for high-dimensional features, the convergence rate is extremely slow. In contrast, by leveraging the DSRS where modality alignment is explicitly constrained by the General Knowledge Loss, PLANET accelerates the convergence rate to $O(C/\sqrt{N})$. Simultaneously, Minimizing $\mathcal{L}_{VQ}$ in Eq. (6) effectively minimizes the upper bounds of $W_1(\hat{\mu}_m, \hat{\nu}_m)$ and $W_1(\hat{\nu}_t, \hat{\mu}_t)$, ensuring that continuous features lie close to DSRS. These mechanisms achieve effective modality alignment.

E. Experiment Settings

E.1. Supervised Learning

Baseline Settings. For supervised models (i.e., GCN, MMGCN, MGAT), we train them directly on downstream datasets without any pre-training. In contrast, for self-supervised and graph foundation models, we follow the pre-training and fine-tuning paradigm: models are first pre-trained across various datasets and subsequently fine-tuned on each specific downstream task.

During the fine-tuning stage, for baselines that do not provide specific methods for node classification or link prediction, we add an MLP to perform these tasks. Specifically, the input embeddings of MLP are encoded by the pretrained graph encoder. For node classification, we set the output dimension of the MLP to the number of classes. For link prediction, we concatenate the embeddings of the target node pair as input to the MLP, where the output dimension is 1, reflecting the score for predicting the pair as a positive sample.

Feature Encoding. Both raw text and image data are encoded using Qwen2-VL-7B-Instruct ([Wang et al., 2024a](#)) to generate high-quality initial node embeddings. For baselines that are incapable of processing MAGs (e.g., GraphMAE2, GFT), the embeddings from modality-specific encoders are concatenated along the feature dimension. It is worth noting that while RiemannGFM and SAMGPT prioritize the learning of topological knowledge, they use node features in specific ways (e.g.,

Table 6. Detailed parameter settings for few-shot node classification and link classification tasks.

Task	Parameter	Value
Few-shot Node Classification	n_{train}	20
	n_{query}	10
	n_{task}	10
Few-shot Link Classification	n_{train}	20
	n_{query}	40
	n_{task}	10

RiemannGFM uses node features in downstream tasks). In our implementation, we replace these node features with the representations encoded by the modality-specific encoders, which have significantly richer semantic information.

Pre-training Datasets and Sampling Strategy. Self-supervised and foundation models are pre-trained in same weights of datasets (weights are presented in Table 5). For baselines that are incapable of taking multiple datasets as input or cannot scale to large-scale graphs, we implement a sampling approach. Specifically, for each node in the graph, we extract its 2-hop neighbor subgraph. We then randomly sample a fixed number of these subgraphs from each dataset according to the pre-defined weights (Table 5), forming the training batches. This guarantees that all models are pre-trained on an identical data distribution.

E.2. Few-Shot Learning

Overall Settings. Most of our experimental settings are the same as those in Appendix E.1 (e.g., feature encoding, sampling strategies.) Following Wang et al. (2024b), we adapt the "Pre-training and Fine-tuning" paradigm. However, during the fine-tuning stage, we do not use an MLP as the classification head. Instead, we use a prototype-based method for classification. The specific values for the parameters used in our experiments are detailed in Table 6.

Few-Shot Node Classification. For an N -way K -shot task, we randomly sample n_{train} samples for each class from the training set. On the test set, we randomly select N classes and sample K instances per class to serve as prototype vectors, along with n_{query} instances for evaluation. These $N \cdot K$ samples are used to construct a prototype classifier, where the embedding of each class is computed by averaging its corresponding K sample embeddings, resulting in N class embeddings. For evaluation, we calculate the cosine similarity between the evaluation vector and each class embedding, assigning the sample to the class with the highest similarity. The same procedure applies to the validation set. To eliminate randomness, this sampling process is repeated n_{task} times, with the final result being the average of these runs.

Few-Shot Link Classification. Since the original datasets (i.e., Amazon-Sports, Amazon-Cloth) are designed for link prediction tasks, we adapt them for link classification tasks. Specifically, we aggregate all positive edges from the training, validation, and test sets to form positive samples. We simultaneously generate an equivalent number of random negative edges to serve as negative samples. These negative edges are allocated to the training, validation, and test sets to strictly match the count of positive edges in each respective set (e.g., a test set containing 500 positive edges is assigned 500 negative edges). Consequently, we formulate the problem as a balanced 2-way K -shot binary classification task. The subsequent method follows the **Few-shot Node Classification** described above.

E.3. Multimodal Generative Tasks

G2Text. We evaluate the performance of G2Text generation on two multimodal datasets:

- **Grocery.** This is a real-world e-commerce dataset. The objective is to generate a comprehensive product description for a target node. The input consists of the target node’s product title and the multimodal context (images and texts) from its neighbors (including target node’s image).
- **Flickr30k.** Originally an image-text retrieval dataset, we adapt it for graph-based generation by constructing a k-NN graph based on feature similarity. Each node contains an image and five distinct textual descriptions. For the G2Text task, we utilize the last four descriptions of the target node and the multimodal information of neighbor nodes to generate the first description.

Table 7. Specific prompts for Grocery and Flickr30k datasets in G2Text task.

Dataset	Prompt
Grocery	### Task: Generate a natural-language description of the product node. ### Input: Title: <i><item title of the target node ></i>. ### Output Results:
Flickr30k	### Task: Generate a detailed description for the image based on the context. ### Input: Context: <i><desc 2 ></i>, <i><desc 3 ></i>, <i><desc 4 ></i>, <i><desc 5 ></i>. ### Output Results:

For the specific prompts used for each dataset, please refer to Table 7.

G2Image. We evaluate the G2Image task on two datasets: Goodreads-NC and Ele-fashion. Following the experimental setting of InstructG2I (Jin et al., 2024), we focus on specific, visually distinct categories to verify whether our embeddings capture sufficient semantic nuances to drive fine-grained generation. For Goodreads-NC, we select nodes belonging to the history and children categories for training and testing. For Ele-fashion, we select nodes from the jewelry and shoes categories. Distinct from the original data partition used in InstructG2I, we adopt a standard randomized partition strategy for our experiments. Specifically, for each selected category subset, we split the data into a training set (80%) and a testing set (20%). This ratio ensures that the model is trained on sufficient data while reserving a substantial portion for robust evaluation.

F. Implementation Notes

Detailed hyper-parameter settings for the pre-training stage are summarized in Table 8. We optimize the entire framework using AdamW optimizer for 5 epochs.

G. Additional Experiments

G.1. Impact of Pre-trained Encoder Choices

Our framework relies on frozen pre-trained encoders to obtain initial modality embeddings before multimodal graph pre-training. To further examine whether the final performance of PLANET is overly dependent on a specific encoder choice, we conduct additional experiments by replacing the original encoders with BERT for text encoding and ViT for image encoding. This setting provides a more challenging test case, since BERT and ViT are independently pre-trained on different objectives and are not explicitly aligned in a shared multimodal semantic space.

The results are shown in Table 9. PLANET is more robust to different choices of pre-trained encoders than UniGraph2. After replacing the original encoders with less-aligned BERT/ViT encoders, PLANET shows a smaller performance drop. This indicates that PLANET relies less on the initial encoder space. This robustness mainly comes from the explicit modality alignment mechanism in PLANET.

G.2. OOD Generalization Evaluation

A foundation model should have strong transferability to unseen graphs and unseen domains. To further evaluate this ability, we conduct additional out-of-domain (OOD) experiments. We follow the OOD setting and definition used in UniGraph2 (He et al., 2025b). The evaluation datasets are divided into three groups. The first group is in-distribution datasets. These datasets are used during pre-training and also used for evaluation. The second group is in-domain generalization datasets. These datasets are not used during pre-training, but they belong to the same domain as one of the pre-training datasets. The third group is OOD generalization datasets. These datasets come from domains that are not seen during pre-training.

The results are shown in Table 10 and Table 11. PLANET achieves the best performance on all in-distribution, in-domain, and out-of-domain evaluation datasets. This result shows that PLANET is not limited to transductive inference on pre-training graphs. PLANET can also transfer to unseen graphs and unseen domains. The improvement mainly comes from its divide-and-conquer design. EDG and NDR modules help PLANET learn transferable multimodal graph representations.

Table 8. Detailed hyper-parameter settings for pre-training.

Category	Parameter	Value
Optimization	learning rate	$4e - 5$
	weight decay	$5e - 4$
	batch size	128
	epochs	5
	optimizer	AdamW
EDG Module	num_experts	5
	top- k	2
	num_layers	8
	num_heads	8
NDR Module	DSRS_size	20480
	DSRS_dim	768
	DSRS_temperature	0.93
Loss	β_1 (feature reconstruction)	0.1
	β_2 (structure reconstruction)	0.1
	β_3 (general knowledge)	0.2
	β_4 (VQ)	0.1
	β_5 (MoE load balance)	0.01
	β_{inter}	0.5
Other Params	hidden_dim (d)	768
	dropout	0.1
	select_node_mask_p	0.6
	select_modality_mask_p	0.4
	edge_mask_p	0.15

Table 9. Impact of different pre-trained encoder choices.

Model / Dataset	Grocery	RedditS	Amazon-Cloth
UniGraph2 (BERT/ViT)	76.91	92.20	19.04
PLANET (BERT/ViT)	83.37	95.41	20.21
UniGraph2 Performance Drop (%)	-6.32%	-1.55%	-1.40%
PLANET Performance Drop (%)	-2.10%	-1.25%	-0.20%

Table 10. OOD generalization results on node classification and link prediction tasks.

Method	Goodreads-NC (In-dis.)	RedditS (In-domain)	Movies (OOD)	Toys (OOD)	Grocery (OOD)	Ele-fashion (OOD)	Amazon-Sports (OOD)
GraphMAE2	72.51	92.75	48.77	75.47	75.33	82.10	24.46
RiemannGFM	78.37	92.04	52.83	78.09	82.17	86.19	22.17
SAMGPT	74.75	92.86	50.28	73.18	75.44	83.95	22.97
UniGraph2	78.55	93.40	52.35	78.73	81.53	86.09	26.55
PLANET	83.81	96.55	56.91	80.36	82.64	87.29	27.63

Table 11. Few-shot OOD generalization results.

Method	Movies 5-way 3-shot	Movies 5-way 5-shot	Amazon-Cloth 2-way 3-shot	Amazon-Cloth 2-way 5-shot
GraphMAE2	44.83	46.25	61.13	61.60
RiemannGFM	54.18	55.70	64.19	64.90
SAMGPT	49.26	56.47	70.85	74.62
UniGraph2	53.40	55.26	71.65	72.78
PLANET	56.67	59.82	73.12	75.36